

Machine Learning Assignment II

Part I: Revised Theoretical Questions

Answer all questions clearly. For numerical questions, show all formulas, intermediate calculations, and final interpretations. Answers without reasoning will not receive full credit.

1. Classification-tree impurity criteria

When constructing a decision tree for classification, criteria such as Gini impurity and cross-entropy are often preferred over raw misclassification error during split selection.

- (a) Define misclassification error, Gini impurity, and cross-entropy for a node m with class proportions $p_{m1}, p_{m2}, \dots, p_{mK}$.
- (b) Explain why Gini impurity and cross-entropy are more sensitive to improvements in node purity than misclassification error, especially when the majority class does not change.
- (c) Give a small binary-class example in which a proposed split improves the class distribution of the child nodes but does not change the majority class. Explain why misclassification error may fail to reward this split while Gini or entropy can reward it.

2. Regression trees, split objective, and geometry

Consider a CART regression tree trained to predict a continuous response Y from continuous predictors $X_1, \dots, X_p$ using mean squared error as the impurity measure.

- (a) Write the mathematical objective used to evaluate a split of node R_m into

$$R_{\text{left}}(j, s) = \{X \in R_m : X_j \leq s\}$$

and

$$R_{\text{right}}(j, s) = \{X \in R_m : X_j > s\}.$$

State what value is predicted in each terminal leaf.

- (b) Suppose a terminal leaf contains target values $[2, 4, 4, 4, 6]$. What numerical prediction is returned for a new observation falling into this leaf? Justify your answer using the MSE objective.
- (c) Describe the geometry of the regions produced by a standard regression tree with continuous predictors. Contrast this with the prediction surface of a multiple linear regression model

$$Y = \beta_0 + \beta_1 X_1 + \cdots + \beta_p X_p.$$

- (d) Discuss how `max_depth` and `min_samples_leaf` influence the bias-variance trade-off.

3. Random Forest versus bagged decision trees

A dataset contains 100 predictors. Only 2 predictors are very strong, while the remaining 98 predictors are weak or noisy. A standard bagged tree ensemble and a Random Forest are both trained on this dataset.

- (a) Explain why trees in a standard bagged ensemble may become highly correlated in this setting.
- (b) Explain how random feature selection at each split can decorrelate trees in a Random Forest.
- (c) The variance of the average of B identically distributed trees with individual variance σ^2 and pairwise correlation ρ can be written as

$$\text{Var}(\bar{f}) = \rho\sigma^2 + \frac{1-\rho}{B}\sigma^2.$$

Interpret this formula and explain why reducing ρ is important.

- (d) State one situation in which random feature selection may hurt predictive performance if the number of candidate features considered at each split is chosen too small.

4. Out-of-bag error and its limitations

Random Forests are often evaluated using out-of-bag (OOB) error.

- (a) Explain how OOB predictions are generated for a training observation.
- (b) Why can OOB error provide an internal estimate of generalization error without a separate validation set?

- (c) In a highly imbalanced binary classification problem, why might overall OOB accuracy be misleading? Explain what additional OOB-based quantities should be inspected for the minority class.
- (d) How would you modify the training or evaluation procedure if the minority class has very few examples?

5. Generative versus discriminative classification

You are building a spam filter to classify emails as Spam or Not Spam based on the words they contain.

- (a) Compare a generative model such as Naive Bayes with a discriminative model such as Logistic Regression. Which model directly estimates $P(Y | X)$, and which model estimates $P(X | Y)$ together with $P(Y)$?
- (b) State the conditional independence assumption made by Naive Bayes. Give a realistic example from text data showing why this assumption is usually violated.
- (c) Despite this unrealistic assumption, explain why Naive Bayes can still perform well in high-dimensional sparse text classification problems.
- (d) If the spam class is rare, explain why accuracy alone is not sufficient and identify at least two more informative evaluation metrics.

6. Deep evaluation of imbalanced classification

A mental-health screening classifier is evaluated on a test set of 1000 individuals. The positive class is `Depression = 1`. The confusion matrix for one decision threshold is:

$$TP = 36, \quad FN = 14, \quad FP = 90, \quad TN = 860.$$

- (a) Compute accuracy, precision, recall/sensitivity, specificity, and F1-score for the positive class.
- (b) A `DummyClassifier` that always predicts `Depression = 0` obtains 95% accuracy on the same test set. Explain why this baseline accuracy is not evidence of a useful screening model.
- (c) For this mental-health problem, discuss whether recall for the Depression class should be prioritized over overall accuracy. Include the consequences of false negatives and false positives.
- (d) Explain how changing the probability threshold affects precision and recall. Which curve, ROC or precision-recall, is usually more informative under severe class imbalance? Justify your answer.

- (e) Suggest two methods for addressing class imbalance during training or model selection, such as class weighting, resampling, threshold tuning, or cost-sensitive learning. Discuss one risk of each method.

7. Numerical classification using Information Gain and Gain Ratio

The following dataset has a binary target class $Y \in \{\text{Yes}, \text{No}\}$. At the root node, compare two candidate categorical splitting attributes, Feature A and Feature B.

Sample	Feature A	Feature B	Y
1	A1	B1	Yes
2	A1	B1	Yes
3	A2	B1	Yes
4	A2	B1	Yes
5	A3	B1	Yes
6	A3	B1	No
7	A4	B2	No
8	A4	B2	No
9	A5	B2	No
10	A5	B2	No
11	A6	B2	Yes
12	A6	B2	No

- Compute the entropy $H(Y)$ at the root node.
- For Feature A, compute the weighted child entropy, Information Gain, Split Information, and Gain Ratio.
- For Feature B, compute the weighted child entropy, Information Gain, Split Information, and Gain Ratio.
- Which feature would be selected by Information Gain? Which feature would be selected by Gain Ratio? Explain why the two criteria can prefer different attributes.
- Discuss why Gain Ratio was introduced to reduce the bias of Information Gain toward attributes with many distinct values.

8. Label noise, loss functions, and robustness

In an image classification task, some dog images are incorrectly labeled as cat and some cat images are incorrectly labeled as dog.

- (a) Compare 0–1 loss with log loss/cross-entropy loss. Why is a smooth probabilistic loss more useful for gradient-based learning than 0–1 loss?
- (b) Explain how mislabeled examples can affect a flexible classifier such as a deep tree or a boosting model. Why can boosting be especially sensitive to hard or mislabeled observations?
- (c) Name three practical strategies for reducing the harm of label noise before or during training.
- (d) Explain why simply removing all difficult examples may introduce bias into the dataset.

9. AdaBoost, XGBoost, and CatBoost

Answer the following questions to compare three boosting algorithms: AdaBoost, XGBoost, and CatBoost.

- (a) Describe the core training mechanism of AdaBoost. How are sample weights updated after each weak learner?
- (b) Describe the core idea of gradient boosting and explain how XGBoost extends it using regularization, shrinkage/learning rate, subsampling, and second-order optimization.
- (c) Explain why CatBoost is designed to handle categorical variables effectively. What is the intuition behind ordered target statistics or ordered boosting in reducing target leakage?
- (d) Compare the three algorithms in terms of sensitivity to noisy labels, ability to handle mixed numerical/categorical data, risk of overfitting, and computational considerations.
- (e) For an imbalanced depression-classification dataset with many categorical predictors, propose a fair experimental comparison among AdaBoost, XGBoost, and CatBoost. Specify the evaluation metrics, cross-validation strategy, and hyperparameters you would tune.

10. Model selection under asymmetric error costs

A hospital research team is comparing Logistic Regression, Random Forest, XGBoost, and a linear SVM for a binary screening task. Missing a positive case is considered more serious than incorrectly flagging a negative case, but false alarms also have clinical and operational costs.

- (a) Explain why choosing the final model only by test accuracy is inappropriate in this setting.
- (b) Define a model-selection strategy that uses cross-validation, multiple metrics, and threshold tuning while avoiding data leakage.
- (c) Discuss how probability calibration may affect threshold-based decisions for models such as Random Forests and boosting models.
- (d) If two models have similar F1-scores but one has much higher recall and lower precision, explain how the final choice should be justified in a formal report.

Part II: Practical Assignment

Practical Assignment: Exploring Mental Health Data

This component requires implementing a complete machine learning pipeline on a mental health survey dataset for binary classification. Your work must reflect graduate-level experimental discipline, including principled preprocessing, strict avoidance of data leakage, and rigorous evaluation of classification performance.

Dataset

Use the dataset containing information about individuals' demographic characteristics, academic or professional status, lifestyle habits, stress levels, satisfaction levels, and mental-health-related background. All preprocessing decisions, missing-value handling, categorical encoding, and feature scaling must be explicitly documented.

Phase I: Exploratory Data Analysis (EDA) and Data Cleaning

Your first objective is to characterize the dataset and extract behavioral patterns that motivate modeling decisions. EDA should quantify dataset scale and describe demographic and lifestyle heterogeneity. Visualizations must be clearly labeled and interpreted; plots presented without analytical conclusions will not receive full credit. Data quality issues, including missingness, inconsistent values, anomalies, and outliers, must be assessed and handled transparently.

EDA must include descriptive statistics for all observations and features, as well as data type identification for each column. You must analyze the distribution of the **Depression** variable, discuss class imbalance, and explain why accuracy alone is insufficient. Numerical variables, such as **Age**, **Academic Pressure**, **Work Pressure**, **CGPA**, and **Study Satisfaction**, should be analyzed in terms of central tendency, variability, and distribution using histograms and boxplots.

You must compare numerical variables across the two depression classes using boxplots or density plots. Categorical variables should be analyzed through frequency distributions, with rare categories identified and discussed. Missing-value analysis must identify which variables contain missing values and document the patterns of missingness. Correlation analysis should be performed for numerical features where appropriate, and visualizations should compare depression rates across categorical variables.

Phase II: Data Preprocessing and Feature Engineering

In this phase, you will prepare the dataset carefully for modeling. Remove non-informative columns such as `id` and `Name`. Handle missing values using appropriate imputation strategies. The dataset contains conditional missingness: students may lack work-related variables, while professionals may lack academic variables. Therefore, you must impute within subgroups based on `Working Professional` or `Student` status, or create indicator variables to preserve missingness information. The chosen method must be justified based on the meaning of the variables and the amount of missing data.

Categorical variables must be encoded before they can be used in most machine learning models. Use one-hot encoding for nominal categorical variables and other non-ordinal categorical features, or use binary encoding for `Working Professional` or `Student`. If any ordinal encoding is used, the ordering must be clearly justified.

Phase III: Predictive Modeling Using Classification

Split the dataset into training and test sets before model training. Use cross-validation on the training set for model selection and hyperparameter tuning. During cross-validation, all preprocessing operations must be performed inside each fold. This means that the imputer, encoder, and scaler must be fitted only on the training portion of each fold and then applied to the validation portion.

You must report evaluation metrics beyond accuracy, including precision, recall, F1-score, ROC-AUC, and confusion matrix. In this context, recall for the `Depression` class must be explicitly discussed.

Depression prediction is formulated as a binary classification task where the target is `Depression`. You must implement and compare multiple classification models. Modeling must focus on a variety of supervised learning approaches, and you must implement and compare at least three of the following model families:

- Logistic Regression
- Linear Discriminant Analysis (LDA) or Quadratic Discriminant Analysis (QDA)
- Decision Tree Classifier
- Tree-based ensemble models: Random Forest and at least one boosting model, such as XGBoost or LightGBM
- Support Vector Machine: Linear SVM and Kernel SVM

Baseline Model Requirement

You must include a simple baseline classifier, such as `DummyClassifier` using the `most_frequent` strategy, that predicts the majority class. This establishes a minimum performance threshold.

Modeling Constraints and Requirements

Use cross-validation with a minimum of 3 folds for hyperparameter tuning. Perform systematic hyperparameter tuning for each model. Report computational considerations, such as training time, and comment on trade-offs. Address class imbalance using appropriate strategies such as class weights, sampling, or threshold adjustment.

For each model, the following evaluation metrics must be reported and interpreted: accuracy, precision, recall, F1-score, ROC-AUC, and confusion matrix. Your analysis must address class imbalance, threshold selection strategy, and precision–recall trade-offs. Model selection must be justified empirically and discussed critically, rather than being based on a single metric.

Phase IV: Model Comparison and Final Model Selection

Compare all selected models in a single model comparison table. Include all models you chose, with a minimum of three models. For each model, report cross-validation performance. The comparison should include accuracy, precision, recall, F1-score, ROC-AUC, and training time.

You should discuss whether recall for the depression class should be prioritized over overall accuracy. In a mental-health-related prediction problem, false negatives may be more concerning because they correspond to depressed individuals who are not identified by the model. However, false positives also have consequences, so the trade-off between precision and recall must be discussed.

For the final selected model, provide a confusion matrix, classification report, and ROC curve. Interpret the model’s strengths and weaknesses based on these results.

Phase V: Model Interpretation and Discussion

Interpret the final model and discuss the most important predictors of depression. Compare interpretations across different model families. For example, discuss whether Logistic Regression coefficients and Random Forest feature importance scores highlight similar variables. If the models disagree, explain why this may happen. Linear models capture additive

relationships, while tree-based models can capture nonlinear effects and interactions between variables.

You must also discuss the limitations of the analysis. These limitations may include missing values, noisy categorical variables, class imbalance, potential survey bias, and the fact that predictive association does not imply causation. The model should not be interpreted as a clinical diagnostic tool. It is a machine learning classifier trained on the available dataset.

Experimental Rigor and Reproducibility

All results must be reproducible. Document data-processing steps and structure the notebook as a clear pipeline: data loading, cleaning, preprocessing, splitting, cross-validation, modeling, and evaluation. The notebook should be readable as a scientific workflow.

Deliverables

Submit a single Jupyter Notebook `.ipynb` containing all code, outputs, and explanatory markdown. Also submit a technical report in PDF format written in formal academic style. The report must include an introduction, dataset description, EDA findings, preprocessing methodology, model training results, hyperparameter tuning, model comparison, final model selection, interpretation of the final model, and discussion of limitations and potential improvements.